Nguyen Minh Hoang

Mechatronics & Industrial Automation Systems

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EDUCATION

Bachelor of Mechatronics Engineering *Graduated November 2026*
Hochiminh City University of Technology *GPA: 3.2 / 4.0*
Relevant Coursework: Robotic System Design, PLC System in Robotics and Automation, Servo System Control, Introduction to Modern Control Systems, Dynamics and Control of Robotic Systems

TECHNICAL SKILLS

- **Programming & Simulation:** Python, MATLAB, Simulink, Unity (Motion Sim).
- **CAD & Prototyping:** SolidWorks (Design and FEA Simulation), AutoCAD.
- **Embedded Systems & Electronics:** ESP32, Raspberry Pi 5.
- **Control & Robotics:** Kinematics & Dynamics, PLC (TIA Portal, GX Works).
- **Networking:** Linux (Debian, Ubuntu, Fedora), Managed/Unmanaged L2/L3 Ethernet & SFP Switching for headless Raspberry Pi nodes and local networked storage.
- **Industrial Maintenance:** Industrial Servo Systems (ESTUN, Omron, Panasonic, LS); Yaskawa VFDs Commissioning & Parameterization (J1000, V1000, GA500).

EXPERIENCE

Mechatronics Engineering Intern *Jun 2024 – Aug 2026 (Seasonal / Recurring Intern, 9 mos total).*
HoangQuan Glass *Binh Chanh, HCM City*

- Assisted in operating and repairing Dual-Edge Glass Grinder as well as Tempered Glass Autoclave; including pneumatic circuits, electrical switchgear, and PLC troubleshooting (Flex 6 Nano, S7-1200 1214C).
- Sourced and programmed industrial controllers (VFDs, PLCs) and electrical switchgear (contactors, thermal overload relays, solid-state relays) from local as well as Chinese suppliers.
- Troubleshot I/O fieldbus communication and calibrated sensors (Panasonic FX-301P and 501P, various other distance sensors) during machine commissioning.
- Sourced and secured VFDs and motors across channels, reducing replacement cost up to 75% for one GA500 ($320/unit) and securing new factory components below secondary-market prices.
- Programmed, commissioned and validated four Yaskawa VFDs (two GA500s, one V1000 and J1000) and Servo systems (three Panasonic MINAS A6, and two LS L7NA). Sourced and worked with third party technicians to repair and replace one ESTUN Pronet-10D system from China.

KEY PROJECTS

Design of a 2-Axis Cartesian Gantry-based Automatic Paint-Spraying Machine *[Document / Video]*
Unity - Snap7 Python - TIA Portal Pipeline, SolidWorks *Jun 2026 – Oct 2026*

- Implemented real-time digital twin and kinematic simulation using Unity - Snap7 - TIA Portal pipeline.
- Commissioned hardware-in-the-loop (HIL) automation using a physical Siemens S7-1200 PLC and Panasonic MINAS A6 servo, synchronizing physical servo axis movement with a real-time Unity digital twin via Snap7 industrial Ethernet communication.

AI-Assisted Sliding Mode Control (Simulation & Stability Analysis) *[GitHub]*
MATLAB, Simulink *Mar 2026 – Apr 2026*

- Trained a TD3 reinforcement learning agent to adapt sliding mode controller parameters in real time (50 Hz) on a 4th-order mass-spring-damper system, replacing fixed-gain control with a learned supervisory policy.
- Established stability of the adaptive control system through dwell-time switching analysis, maintaining a 16.9 ms switching interval within the 20 ms control-loop constraint.
- Cut peak actuator force by 51% (±1,196 N vs. ±2,427 N) with no increase in power draw — extending projected actuator lifespan versus fixed-gain SMC.
- Conducted 100-run Monte Carlo evaluations across multiple seeds, identifying late-stage policy drift and benchmarking against seven control topologies (MPC, LQR, Adaptive SMC).